Article

Rigidity at the Extremal Crouzeix Constant

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Abstract

For a complex matrix A, let $\psi(A)$ be the norm of the polynomial functional calculus on its numerical range. We prove that $\psi(A) = 2$ only when the numerical range is a nondegenerate closed disk. The proof first extracts the equality conditions in the sharp double-layer estimate. After boundary eigenvalues are removed, a finite-Blaschke extremizer satisfies an exact boundary-kernel identity. On a curved exposed arc this identity makes the extremizer rational. An extremal state then places every zero of its reduced numerator inside the numerical range and every pole outside; consequently its full unit lemniscate coincides with the boundary of the numerical range. The dual of that algebraic boundary divides the Hermitian Kippenhahn determinant. Hyperbolicity and strict convexity force the dual curve to have degree two, and its points at infinity force the resulting conic to be a circle.

Keywords: numerical range; Crouzeix ratio; spectral set; rational lemniscate; hyperbolic polynomial

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1. Introduction

For $A \in M_N(\mathbb{C})$, set

$$W(A) = \{u^*Au : u \in \mathbb{C}^N,\ \|u\|_2 = 1\}.$$

For a polynomial p, let $\|p(A)\|$ be its Euclidean operator norm and put

$$\psi(A) = \sup\left\{\|p(A)\| : p \in \mathbb{C}[z],\ \max_{z\in W(A)} |p(z)| \le 1\right\}.$$

This quantity was introduced by Crouzeix in connection with the conjecture that $\psi(A) \le 2$ for every matrix [1]. The integral representation behind the modern estimates goes back to Delyon and Delyon [2]. Crouzeix and Palencia proved the bound $1 + \sqrt{2}$ [3], and Lorist and Schwenninger established the sharp bound 2 [4]. We retain the equality information in the double-layer and iterate arguments. The resulting rigidity statement is the following.

Theorem 1. *For every $N \ge 1$ and $A \in M_N(\mathbb{C})$,*

$$\psi(A) = 2 \quad \Longrightarrow \quad W(A) = \{z \in \mathbb{C} : |z - \gamma| \le \rho\}$$

for some $\gamma \in \mathbb{C}$ and some $\rho > 0$.

The proof has four steps. An equality analysis of the sharp bound produces two extremal vectors and a pointwise boundary-kernel identity. A curved exposed arc then turns that identity into a rational formula for the extremizer. A positive state identifies its full unit lemniscate with $\partial W(A)$. Finally, the Kippenhahn determinant converts the tangent curve into a hyperbolic factor; a generic contact count shows that this factor is a conic, hence a circle. All scalar matrix coefficients are written as x^*By.

2. Numerical-range preliminaries

Lemma 1 (Numerical-range geometry)**.** *The following facts hold.*

1. *$W(A)$ is nonempty, compact, and convex.*
2. *If $W(A)$ is a singleton or is contained in a line, then A is normal and $\psi(A) = 1$.*
3. *For $a \neq 0$, $b \in \mathbb{C}$, and unitary $\mathcal{U}$,*

$$W(aA + bI) = aW(A) + b, \qquad \psi(aA + bI) = \psi(A), \qquad \psi(\mathcal{U}^* A \mathcal{U}) = \psi(A).$$

Also $W(A^*) = \overline{W(A)}$ and $\psi(A^*) = \psi(A)$.

4. *For a finite direct sum,*

$$W\left(\bigoplus_j A_j\right) = \operatorname{conv}\Big(\bigcup_j W(A_j)\Big), \qquad \psi\left(\bigoplus_j A_j\right) \leq \max_j \psi(A_j).$$

5. *The support function of $W(A)$ is*

$$h_A(\theta) = \lambda_{\max}\Big(\operatorname{Re}(e^{-i\theta} A)\Big).$$

Proof. The unit sphere is compact and $u \mapsto u^* A u$ is continuous, proving nonemptiness and compactness. To prove convexity, take two points of $W(A)$ and compress A to the span of corresponding unit vectors. It is enough to know that the numerical range of a 2×2 matrix is convex. After unitary triangularization such a matrix has the form $\left(\begin{smallmatrix} \lambda_1 & c \\ 0 & \lambda_2 \end{smallmatrix}\right)$. For $u = (\cos t, e^{i\varphi} \sin t)$,

$$u^* A u = \lambda_1 \cos^2 t + \lambda_2 \sin^2 t + c e^{i\varphi} \sin t \cos t.$$

These points fill the possibly degenerate ellipse with foci λ_1, λ_2, hence a convex set. The two-dimensional compression is contained in $W(A)$, so the segment joining the original two points is contained in $W(A)$.

If $W(A) = \{\gamma\}$, then $x^*(A - \gamma I)x = 0$ for all x. The complex polarization identity makes every matrix coefficient of $A - \gamma I$ zero. If $W(A)$ lies in a line, a translation and a nonzero complex rotation make all quadratic forms of A real. The skew-Hermitian part then has zero quadratic form and vanishes by polarization, so the transformed matrix is Hermitian. A normal matrix satisfies

$$\|p(A)\| = \max_{\lambda \in \operatorname{spec}(A)} |p(\lambda)| \leq \max_{z \in W(A)} |p(z)|.$$

The constant polynomial (1) gives the reverse inequality.

The affine and unitary formulas follow by changing variables in polynomials. For adjoints use $\widetilde{p}(z) = \overline{p(\overline{z})}$, for which $\widetilde{p}(A^*) = p(A)^*$. For a direct sum, a unit vector decomposes into block components and its numerical value is a convex combination of block numerical values. Also $p(\oplus A_j) = \oplus p(A_j)$; the supremum norm on the convex hull dominates that on every block. Finally,

$$\max_{z \in W(A)} \operatorname{Re}(e^{-i\theta} z) = \max_{\|u\|=1} u^* \operatorname{Re}(e^{-i\theta} A) u,$$

which is the largest eigenvalue of the displayed Hermitian matrix. □

3. The sharp bound and its equality case

We include the sharp estimate because the equality conditions are the starting point of the rigidity argument. The next lemma is the iterated-dilation estimate of Lorist and Schwenninger [4], followed by the equality analysis needed here.

Lemma 2 (An abstract dilation estimate)**.** *Let T act on a finite-dimensional Hilbert space. Suppose that $\mathcal{M}$ is a contraction on another Hilbert space, $\mathcal{V}$ is an isometry, and*

$$E_n = 2\mathcal{V}^*\mathcal{M}^{*n}\mathcal{V} - T^{*n} \qquad (n \geq 1)$$

are uniformly bounded and commute with T. Then $\|T\| \leq 2$.

If additionally $\|T\| = 2$, $r(T) < 1$, $T^*Tx = 4x$, $\|x\| = 1$, and $y = Tx/2$, then

$$T^*x = 0, \quad T^*y = 2x, \quad \mathcal{M}\mathcal{V}x = \mathcal{V}y, \quad \mathcal{M}^*\mathcal{V}y = \mathcal{V}x.$$

Proof. Put $\kappa = \|T\|$. There is nothing to prove if $\kappa \leq 1$. Choose a unit vector x with $T^*Tx = \kappa^2 x$, and set

$$m_n = \operatorname{Re}\big(x^* E_n T^n x\big), \qquad \omega = \mathcal{M}^*\mathcal{V}Tx - \kappa\mathcal{V}x.$$

The defining identity gives $\|T^{*n}\| \leq 2 + \sup_k \|E_k\|$, so the powers of T, and hence (m_n), are bounded. Put $Q_n = \mathcal{V}^*\mathcal{M}^{*n}\mathcal{V}$ and $q_n = T^{*n}x$. Commutativity of E_n with T gives

$$2(Q_{n+1}T - TQ_{n+1}) = T^{*(n+1)}T - TT^{*(n+1)}.$$

Moreover, $m_n = 2\operatorname{Re}(q_n^* Q_n x) - \|q_n\|^2$ and $\mathcal{V}^*\mathcal{M}^{*n}\omega = Q_{n+1}Tx - \kappa Q_n x$. Multiplying the displayed commutator identity on the left by q_n^*, using $T^*Tx = \kappa^2 x$, and taking real parts now gives

$$\begin{aligned}
\kappa m_n - m_{n+1} &= \kappa(\kappa-1)\|T^{*n}x\|^2 - 2\operatorname{Re}\big((\mathcal{V}^*\mathcal{M}^{*n}\omega)^* T^{*n}x\big) \\
&= \kappa(\kappa-1)\left\| T^{*n}x - \frac{\mathcal{V}^*\mathcal{M}^{*n}\omega}{\kappa(\kappa-1)} \right\|^2 - \frac{\|\mathcal{V}^*\mathcal{M}^{*n}\omega\|^2}{\kappa(\kappa-1)} \\
&\geq -\frac{\|\omega\|^2}{\kappa(\kappa-1)}.
\end{aligned}$$

Telescoping, with the boundedness of m_n, yields

$$\kappa m_1 = \sum_{n\geq 1} \kappa^{-n+1}(\kappa m_n - m_{n+1}) \geq -\frac{\|\omega\|^2}{(\kappa-1)^2}. \tag{3.1}$$

On the other hand, using that $\mathcal{V}$ is isometric and $\mathcal{M}$ contractive,

$$\begin{aligned}
\|\omega\|^2 &\leq 2\kappa^2 - 2\kappa\operatorname{Re}\big(x^*\mathcal{V}^*\mathcal{M}^*\mathcal{V}Tx\big) \\
&= 2\kappa^2 - \kappa m_1 - \kappa^3.
\end{aligned} \tag{3.2}$$

Combining (3.1) and (3.2) gives

$$\|\omega\|^2\left(1 - \frac{1}{(\kappa-1)^2}\right) \leq \kappa^2(2-\kappa),$$

which is impossible for $\kappa > 2$.

Now let $\kappa = 2$. Equations (3.1)–(3.2) become $2m_1 \geq -\|\omega\|^2$ and $\|\omega\|^2 \leq -2m_1$, so equality holds in both. For each n, the slack in the estimate preceding (3.1) is

$$2\left\| T^{*n}x - \tfrac{1}{2}\mathcal{V}^*\mathcal{M}^{*n}\omega \right\|^2 + \tfrac{1}{2}\big(\|\omega\|^2 - \|\mathcal{V}^*\mathcal{M}^{*n}\omega\|^2\big).$$

All these nonnegative slacks have a positive weighted sum equal to zero. Thus, for every (n),

$$\mathcal{V}^*\mathcal{M}^{*n}\omega = 2T^{*n}x, \qquad \|\mathcal{V}^*\mathcal{M}^{*n}\omega\| = \|\omega\|.$$

Since $r(T) < 1$, $T^{*n}x \to 0$, and these two equalities force $\omega = 0$. Their first instance then gives $T^*x = 0$. With $y = Tx/2$, $\mathcal{M}^*\mathcal{V}y = \mathcal{V}x$. Both $\mathcal{V}x$ and $\mathcal{V}y$ are unit vectors, so equality in Cauchy–Schwarz for $\mathcal{M}$ gives $\mathcal{M}\mathcal{V}x = \mathcal{V}y$. Finally $T^*y = T^*Tx/2 = 2x$. $\square$

Proposition 1 (The sharp numerical-range estimate)**.** *For every finite matrix A and every polynomial p,*

$$\|p(A)\| \le 2 \max_{z \in W(A)} |p(z)|. \tag{3.3}$$

More generally, if K is a compact convex body, $W(A) \subset K$, $\operatorname{spec}(A) \subset \operatorname{int} K$, and $f \in A(K)$, then

$$\|f(A)\| \le 2\|f\|_K. \tag{3.4}$$

The same holds for $f \in H^\infty(\operatorname{int} K)$.

Proof. Normalize $\|f\|_K \le 1$. Cauchy's boundary formula below also holds when f is only in $A(K)$: for $z_0 \in \operatorname{int} K$, the functions $f(z_0 + r(z - z_0))$, $r < 1$, are holomorphic on a neighborhood of K and converge uniformly to f; apply the usual formula to them and pass to the limit. This is the positive double-layer construction used in [2,3]. Parametrize the rectifiable convex Jordan curve $\Gamma = \partial K$ counterclockwise by arclength. Its outward unit normal $\mathfrak{n}(\sigma)$ exists almost everywhere. Put $R_\sigma = (\sigma I - A)^{-1}$ and

$$\mathcal{P}(\sigma) = \frac{1}{\pi} \operatorname{Re}\big(\mathfrak{n}(\sigma) R_\sigma\big).$$

At a point with a normal, the supporting-half-plane property gives

$$\Delta_\sigma := \operatorname{Re}\big(\overline{\mathfrak{n}(\sigma)}(\sigma I - A)\big) \succeq 0.$$

Multiplication by the two resolvents shows

$$\operatorname{Re}(\mathfrak{n} R_\sigma) = R_\sigma^* \Delta_\sigma R_\sigma \succeq 0. \tag{3.5}$$

Since $d\sigma = i\mathfrak{n}(\sigma)\, ds$, Cauchy's formula and its adjoint give

$$\int_\Gamma \mathcal{P}(\sigma)\, ds = 2I. \tag{3.6}$$

Consequently

$$(\mathcal{V}x)(\sigma) = 2^{-1/2} \mathcal{P}(\sigma)^{1/2} x$$

defines an isometry into $L^2(\Gamma; \mathbb{C}^N)$. Let M_f be multiplication by f, and set $T = f(A)$. Splitting the two terms in $\mathcal{P}$ and using Cauchy's formula gives, for every $n \ge 1$,

$$E_n := 2\mathcal{V}^* M_f^{*n} \mathcal{V} - T^{*n} = \frac{1}{2\pi i} \int_\Gamma \overline{f(\sigma)^n} R_\sigma\, d\sigma. \tag{3.7}$$

The right side commutes with T, because every resolvent does. Also

$$\sup_n \|E_n\| \le \frac{\operatorname{length}(\Gamma)}{2\pi} \max_{\sigma \in \Gamma} \|R_\sigma\| < \infty.$$

Lemma 2 proves (3.4).

For an H^∞ function, choose a Riemann map $\phi : \operatorname{int} K \to \mathbb{D}$ and replace f by

$$f_r(z) = f\big(\phi^{-1}(r\phi(z))\big), \qquad r < 1.$$

These functions belong to $A(K)$, retain norm at most 1, and their spectral jets converge to those of f; hence $f_r(A) \to f(A)$.

Finally, for an arbitrary A, apply (3.4) to the outer parallel convex body $K_\varepsilon = W(A) + \varepsilon\overline{\mathbb{D}}$, which contains $W(A)$ in its interior. For a polynomial p, $\max_{K_\varepsilon} |p| \to \max_{W(A)} |p|$. Letting $\varepsilon \downarrow 0$ proves (3.3), including all boundary-spectrum and nonsmooth cases. □

Proposition 2 (Equality data). *Let $K = W(A)$ have nonempty interior, let $\operatorname{spec}(A) \subset \Omega := \operatorname{int} K$, and let $f \in A(K)$ be nonconstant with*

$$\|f\|_K = 1, \qquad |f| = 1 \text{ on } \partial K, \qquad \|f(A)\| = 2.$$

Then there are orthonormal vectors x, y such that, with $T = f(A)$,

$$Tx = 2y, \quad T^*x = 0, \quad Ty = 0, \quad T^*y = 2x, \tag{3.8}$$

and, for almost every $\sigma \in \partial K$,

$$\mathcal{P}(\sigma)^{1/2}\big(y - f(\sigma)x\big) = 0. \tag{3.9}$$

Moreover, for every function h holomorphic near $\operatorname{spec}(A)$ for which the products below are defined,

$$x^*h(A)f(A)x = 0, \qquad y^*h(A)f(A)x = 2x^*h(A)x, \tag{3.10}$$
$$x^*h(A)y = 0, \qquad y^*h(A)y = x^*h(A)x. \tag{3.11}$$

Proof. Apply the construction in Proposition 1. The spectral mapping theorem gives $r(T) < 1$, because the finitely many eigenvalues of A are inside Ω and a nonconstant inner function has modulus strictly less than 1 there. Choose x with $T^*Tx = 4x$, and put $y = Tx/2$. Lemma 2 gives $T^*x = 0$, $T^*y = 2x$, and $M_f\mathcal{V}x = \mathcal{V}y$. Here $\|y\| = 1$, and $2x^*y = x^*Tx = (T^*x)^*x = 0$, so x, y are orthonormal. The multiplication identity is exactly (3.9).

Apply the same equality argument to A^*, the domain $\overline{K}$, and $\widetilde{f}(z) = \overline{f(\overline{z})}$. Its matrix value is T^*, and y is a top right singular vector for T^*. The conclusion $(T^*)^*y = 0$ is $Ty = 0$. Thus (3.8) holds.

Since T commutes with $h(A)$, $T^*x = 0$ gives the first equality in (3.10), and $Tx = 2y$ then gives the first equality in (3.11). Also

$$x^*h(A)x = \tfrac{1}{2}y^*Th(A)x = \tfrac{1}{2}y^*h(A)Tx = y^*h(A)y,$$

and multiplication by $Tx = 2y$ gives the second equality in (3.10). □

Lemma 3 (Boundary eigenvalues reduce). *If $Ax = \lambda x$, $\|x\| = 1$, and $\lambda \in \partial W(A)$, then $A^*x = \overline{\lambda}x$. Consequently*

$$A \simeq A_\partial \oplus A_{\mathrm{in}},$$

where A_∂ is diagonal and consists of all eigenvalues on $\partial W(A)$, while $\operatorname{spec}(A_{\mathrm{in}}) \subset \operatorname{int} W(A)$.

Proof. Choose a supporting direction n, $|n| = 1$, at λ. Equality in the Rayleigh quotient gives

$$\operatorname{Re}(\overline{n}A)x = \operatorname{Re}(\overline{n}\lambda)x.$$

Substituting $Ax = \lambda x$ gives $A^*x = \overline{\lambda}x$. Thus $\mathbb{C}x$ reduces A. Split it off and repeat. Every eigenvalue belongs to $W(A)$, since an eigenvector realizes it, so the remaining spectrum is in the interior. □

Lemma 4 (Conformal and finite-interpolation facts). *Let Ω be a bounded convex domain.*

1. *There is a conformal bijection $\phi : \Omega \to \mathbb{D}$, and it extends to a homeomorphism $\overline{\Omega} \to \overline{\mathbb{D}}$.*

2. *Given finitely many values and derivatives at points of Ω that are realized by a function of $H^\infty(\Omega)$ of norm at most 1, the same data are realized by $\mathcal{B} \circ \phi$, where $\mathcal{B}$ is a finite Blaschke product. If the least possible norm for nonconstant data is exactly 1, the norm-one interpolant is unique and is of this form.*

Proof. A bounded convex domain is a Jordan domain. The Riemann mapping theorem and the Carathéodory boundary theorem therefore give the first assertion.

Transfer the jet data to $\mathbb{D}$ by ϕ^{-1}. The confluent Nevanlinna–Pick theorem says that such finite data are feasible in the Schur class exactly when their generalized Pick matrix is positive semidefinite. The Schur recursion, with the final free parameter chosen unimodular, then produces a rational inner solution, hence a finite Blaschke product. If c_0 is the least common norm of nonconstant data, the generalized Pick matrix for the data divided by c_0 is singular: positive definiteness would persist after a small decrease of c_0. After redundant jet conditions are consolidated, a singular minimal confluent Pick problem has a unique Schur solution, and the same recursion shows that it is a finite Blaschke product. These finite-interpolation facts, including repeated nodes, are also recorded in the numerical-range setting in [1]. $\square$

For a bounded domain Ω and a matrix A_0 with $\operatorname{spec}(A_0) \subset \Omega$, define

$$\psi_\Omega(A_0) = \sup\{\|g(A_0)\| : g \in H^\infty(\Omega),\ \|g\|_\Omega \leq 1\}.$$

The value $g(A_0)$ depends on only finitely many derivatives, those prescribed by the Jordan blocks. Montel's theorem therefore makes the supremum a maximum. Lemma 4 replaces an extremizer by a finite Blaschke product composed with a Riemann map, without changing $g(A_0)$.

Lemma 5 (Strict domain monotonicity)**.** *Let $\Omega_0 \subsetneq \Omega_1$ be bounded convex domains and suppose* $\operatorname{spec}(A_0) \subset \Omega_0$. *If $\psi_{\Omega_1}(A_0) > 1$, then*

$$\psi_{\Omega_0}(A_0) > \psi_{\Omega_1}(A_0).$$

Proof. Monotonicity in the displayed direction is immediate by restriction. Suppose equality held, with common value $c > 1$, and choose a norm-one extremizer g on Ω_1. Its norm on Ω_0 must be exactly 1, since otherwise rescaling its restriction gives a value larger than c. Thus it is also an extremizer on Ω_0.

For either domain, interpolate the finite jet of g that determines $g(A_0)$ with least possible norm. A norm smaller than 1 would contradict extremality after rescaling. The uniqueness part of Lemma 4 says that g is a finite Blaschke product relative to both domains. Hence it has modulus 1 on each boundary.

Choose $a \in \Omega_0$. Along every ray from a, convexity gives an exit radius for each domain. Since the domains differ, on some ray the first exit occurs strictly earlier; hence there is $\xi \in \partial\Omega_0 \cap \Omega_1$. At this interior point of Ω_1, $|g(\xi)| = 1$. The maximum-modulus principle makes g constant, which would give $c = 1$, a contradiction. $\square$

4. Reduction to an interior-spectrum equality pair

Lemma 6 (Boundary-spectrum induction)**.** *Assume Theorem 1 has been proved in every size smaller than N. Let $A \in M_N(\mathbb{C})$, let $K = W(A)$, and suppose $\psi(A) = 2$. Then either K is a nondegenerate closed disk, or*

$$\operatorname{spec}(A) \subset \operatorname{int} K. \tag{4.1}$$

Proof. By Lemma 1, K has nonempty interior. Choose polynomials p_j such that

$$\max_K |p_j| = 1, \qquad \|p_j(A)\| \longrightarrow 2. \tag{4.2}$$

Suppose A has a boundary eigenvalue. Lemma 3 gives $A \simeq A_\partial \oplus A_1$, where A_∂ is diagonal with spectrum on ∂K and $\operatorname{spec}(A_1) \subset \operatorname{int} K$. The A_∂-block of $p_j(A)$ has norm at most 1, so

$$\|p_j(A_1)\| \longrightarrow 2. \tag{4.3}$$

In particular A_1 is nonempty. Since $W(A_1) \subset K$, (4.3) and Proposition 1 give $\psi(A_1) = 2$. Its size is smaller than N, so induction says that

$$K_1 := W(A_1)$$

is a closed disk of positive radius.

There is one more necessary peeling step. Apply Lemma 3 to A_1 relative to K_1:

$$A_1 \simeq A_{\partial,1} \oplus A_2,$$

where $A_{\partial,1}$ is a finite diagonal block on ∂K_1 and $\operatorname{spec}(A_2) \subset \operatorname{int} K_1$. The same sequence (4.2) is bounded by 1 at the eigenvalues of $A_{\partial,1}$, all of which lie in K, so $\|p_j(A_2)\| \to 2$. Thus A_2 is nonempty.

Write $\Lambda_1 = \operatorname{spec}(A_{\partial,1})$. Since

$$K_1 = W(A_1) = \operatorname{conv}(\Lambda_1 \cup W(A_2)),$$

every extreme point of K_1 belongs to the compact set $\Lambda_1 \cup W(A_2)$. Here is a direct verification of this last elementary fact: if an extreme point of the convex hull of a compact set were not in the set, a finite convex representation with at least two distinct points would write it as a nontrivial segment point. The circle ∂K_1 has infinitely many extreme points and Λ_1 is finite. Hence $\partial K_1 \setminus \Lambda_1 \subset W(A_2)$; closedness and convexity of $W(A_2)$ give

$$W(A_2) = K_1. \tag{4.4}$$

Let $\Omega_0 = \operatorname{int} K_1$ and $\Omega_1 = \operatorname{int} K$. The polynomial sequence and Proposition 1 show

$$\psi_{\Omega_0}(A_2) = \psi_{\Omega_1}(A_2) = 2. \tag{4.5}$$

Indeed the lower bounds follow from (4.2)–(4.4), while the upper bounds are the relative estimate (3.4). If $K_1 \subsetneq K$, then $\Omega_0 \subsetneq \Omega_1$, and Lemma 5 contradicts (4.5). Therefore $K_1 = K$, and K is the asserted disk. If this conclusion does not occur, A has no boundary eigenvalue, which is (4.1). □

Lemma 7 (The holomorphic extremizer)**.** *Suppose $K = W(A)$ has nonempty interior and* $\operatorname{spec}(A) \subset \Omega = \operatorname{int} K$*. If $\psi(A) = 2$, there is a nonconstant*

$$f = \mathcal{B} \circ \phi \in A(K),$$

where $\phi : \Omega \to \mathbb{D}$ is conformal and $\mathcal{B}$ is a finite Blaschke product, such that

$$\|f\|_K = 1, \qquad |f| = 1 \text{ on } \partial K, \qquad \|f(A)\| = 2. \tag{4.6}$$

Proof. Normalized polynomials witnessing $\psi(A) = 2$ show $\psi_\Omega(A) \geq 2$, and Proposition 1 gives the reverse inequality. Montel compactness, followed by convergence of the finitely many spectral jets, gives a norm-one $H^\infty(\Omega)$ maximizer. Apply the finite Schur algorithm in Lemma 4 to the jet that determines its matrix value. It supplies $\mathcal{B} \circ \phi$ with the same matrix value. The boundary extension of ϕ puts this function in $A(K)$ and gives boundary modulus 1. It cannot be constant because a constant matrix has norm at most 1. □

5. From equality to a rational inner function

We first locate an analytic boundary arc.

Lemma 8 (A curved numerical-range arc)**.** *If $K = W(A)$ has interior, $\operatorname{spec}(A) \subset \operatorname{int} K$, and K is not a polygon, then ∂K contains an open real-analytic arc on which every supporting line exposes one point and the curvature is positive. Moreover, under $\operatorname{spec}(A) \subset \operatorname{int} K$, K cannot be a polygon.*

Proof. Put
$$\mathcal{H}(\theta) = \operatorname{Re}(e^{-i\theta}A), \qquad h_A(\theta) = \lambda_{\max}(\mathcal{H}(\theta)).$$
Rellich's analytic perturbation theorem [5] shows that the roots of the characteristic polynomial of the analytic Hermitian pencil $\mathcal{H}(\theta)$ can locally be labeled by real-analytic functions. One direct proof is to split the spectral projections by small Cauchy circles and then diagonalize the resulting analytic Hermitian blocks; repeated identical branches are merged. Two distinct analytic algebraic branches have only finitely many crossings on the circle. Thus the circle has a finite partition into intervals on which h_A is analytic.

At a differentiability point, the exposed boundary point is

$$\sigma(\theta) = e^{i\theta}\big(h_A(\theta) + ih_A'(\theta)\big), \qquad \sigma'(\theta) = ie^{i\theta}\big(h_A(\theta) + h_A''(\theta)\big). \tag{5.1}$$

The first identity follows by differentiating the top Rayleigh quotient; the second follows by differentiation. Convexity gives $h_A + h_A'' \geq 0$. If this function vanished identically on every analytic interval, σ would be constant on each interval and K would be a polygon. Otherwise analyticity supplies a subinterval on which $h_A + h_A'' > 0$. Formula (5.1) then gives the required regular, positively curved arc. Differentiability of the support function makes the exposed face a singleton there.

It remains to exclude a polygon when the spectrum is interior. Let λ be a vertex and let a unit vector v realize $v^*Av = \lambda$. Two linearly independent supporting normals n_1, n_2 at the vertex give

$$\operatorname{Re}(\overline{n_j}A)v = \operatorname{Re}(\overline{n_j}\lambda)v \quad (j = 1, 2).$$

The two independent real-linear equations imply $Av = \lambda v$ and $A^*v = \overline{\lambda}v$. Thus λ is a boundary eigenvalue, contradicting $\operatorname{spec}(A) \subset \operatorname{int} K$. □

Proposition 3 (Rational collapse)**.** *Under the hypotheses and notation of Lemma 7, let x, y be the equality vectors from Proposition 2. Then the function f is rational on Ω. More precisely, if*

$$a(z) = x^*(zI - A)^{-1}x, \qquad c(z) = y^*(zI - A)^{-1}x,$$

then
$$c(z)f(z) = 2a(z) \tag{5.2}$$

as an identity of meromorphic functions on Ω. Equivalently, $f = 2a/c$ wherever the quotient is defined, and then everywhere after removable cancellation. After cancellation, $f = U/V$ for coprime polynomials with

$$\deg U > \deg V. \tag{5.3}$$

Proof. Choose the arc from Lemma 8; it avoids the finite spectrum. At an almost-everywhere point σ of that arc, let $\mathfrak{n}$ be its outward normal, $R = (\sigma I - A)^{-1}$, and

$$\Delta_\sigma = \operatorname{Re}\big(\overline{\mathfrak{n}}(\sigma I - A)\big) \succeq 0.$$

Equations (3.5) and (3.9) imply
$$\Delta_\sigma R\big(y - f(\sigma)x\big) = 0. \tag{5.4}$$

Indeed, squaring the norm in (3.9) and using $\mathcal{P} = \pi^{-1} R^* \Delta_\sigma R$ first gives $\Delta_\sigma^{1/2} R(y - f(\sigma)x) = 0$, and hence (5.4). The vector $r_\sigma = R(y - f(\sigma)x)$ is nonzero: R is invertible, while the orthonormality of x, y makes $y - f(\sigma)x \neq 0$. It is therefore a top support eigenvector. The exposed point is unique, so $r_\sigma^* A r_\sigma = \sigma \|r_\sigma\|^2$. Since $(\sigma I - A) r_\sigma = y - f(\sigma)x$, taking a scalar product and then conjugating gives

$$\big(y - f(\sigma)x\big)^* R\big(y - f(\sigma)x\big) = 0. \tag{5.5}$$

The resolvent commutes with $T = f(A)$. Relations (3.8) give, for every $z \notin \operatorname{spec}(A)$,

$$x^* R_z y = 0, \qquad y^* R_z y = x^* R_z x = a(z). \tag{5.6}$$

Expanding (5.5), using $|f(\sigma)| = 1$ and (5.6), yields

$$2a(\sigma) - f(\sigma)c(\sigma) = 0 \tag{5.7}$$

almost everywhere on the arc. Both sides are continuous there, so (5.7) holds on the whole arc. On a collar of this arc the function $cf - 2a$ is analytic, continuous to the arc, and zero there, because the spectrum is a positive distance away. The arc is regular and real analytic, so a local biholomorphic coordinate flattens it to a line segment; convexity places Ω on one side. Extend $cf - 2a$ by zero to the other side. The extension is continuous, and Morera's theorem makes it analytic. The identity theorem first gives $cf - 2a = 0$ in an interior collar. The punctured domain $\Omega \setminus \operatorname{spec}(A)$ is connected; applying the identity theorem there proves $cf - 2a = 0$ throughout it, hence as a meromorphic identity on Ω. Since $a(z) = z^{-1} + O(z^{-2})$ at infinity, a is not the zero rational function; the identity therefore also shows that c is not identically zero. This proves (5.2).

Write

$$a(z) = \frac{x^* \operatorname{adj}(zI - A)x}{\det(zI - A)}, \qquad c(z) = \frac{y^* \operatorname{adj}(zI - A)x}{\det(zI - A)}.$$

The numerator of a has degree exactly $N - 1$, with leading coefficient 1. Since $x^* y = 0$, the numerator of c has degree at most $N - 2$. Canceling a common polynomial from numerator and denominator preserves the strict degree difference, proving (5.3). □

6. The full lemniscate is the boundary

Lemma 9 (The extremal state)**.** *With the notation above, the functional*

$$L(h) = x^* h(A) x, \qquad h \in A(K),$$

is a unital positive state: if $\operatorname{Re} h \geq 0$ *on K, then* $\operatorname{Re} L(h) \geq 0$*, and* $\|L\| = L(1) = 1$*. Consequently there is a probability measure μ on K such that*

$$L(h) = \int_K h(\zeta)\, d\mu(\zeta) \qquad (h \in A(K)). \tag{6.1}$$

Proof. If $\operatorname{Re} h \geq 0$, then $f_t = f e^{-th}$ has supremum norm at most 1 for $t \geq 0$. Since f is extremal and $Tx = 2y$,

$$\begin{aligned} 0 &\geq \frac{d}{dt}\bigg|_{0+} \|f_t(A)x\|^2 \\ &= -2\operatorname{Re}\big((Tx)^* T h(A) x\big) = -8 \operatorname{Re}\big(y^* h(A) y\big) = -8 \operatorname{Re} L(h), \end{aligned}$$

where the last equality is (3.11). Thus L is positive on real parts and $L(1) = 1$. If $\|h\|_K \leq 1$, apply positivity to $1 - e^{i\theta} h$ for every θ to get $|L(h)| \leq 1$. Hence $\|L\| = 1$.

Extend L from the unital function space $A(K)$ to $C(K)$ by the norm-preserving Hahn–Banach theorem. A norm-one functional taking 1 to 1 is positive: applying its norm bound to $1 + itg$, with g

real and letting $t \to 0$, first makes its value on g real; applying it to $0 \le g \le 1$ then gives a value in $[0, 1]$. The elementary Riesz representation construction (define the mass of an open set by suprema of continuous functions supported in it) gives a probability measure μ and (6.1). $\square$

Lemma 10 (Zeros of the transfer function)**.** *Let $f = U/V$ be the reduced rational function from Proposition 3. Then every finite zero of U lies in Ω, every zero of V lies outside K, and $f(\infty) = \infty$.*

Proof. For $z \notin K$, the function $\zeta \mapsto (z-\zeta)^{-1}$ lies in $A(K)$. Thus (6.1) gives

$$a(z) = \int_K \frac{d\mu(\zeta)}{z-\zeta}. \tag{6.2}$$

Strict separation of z from the convex set K supplies a unit complex number η such that $\operatorname{Re}(\overline{\eta}(z-\zeta)) > 0$ for every $\zeta \in K$. Therefore

$$\operatorname{Re}\big(\eta a(z)\big) = \int_K \operatorname{Re} \frac{\eta}{z-\zeta}\, d\mu(\zeta) > 0$$

because, on writing $z - \zeta = \eta w$, the separation inequality says $\operatorname{Re} w > 0$, and then $\operatorname{Re}(\eta/(z-\zeta)) = \operatorname{Re}(1/w) > 0$. In particular $a(z) \neq 0$ off K.

The reduced numerator U of $f = 2a/c$ divides the numerator of a, so it has no zero off K. It has no zero on ∂K, where $|f| = 1$, and hence all its zeros lie in Ω. A reduced pole cannot lie in Ω, where f is holomorphic, or on ∂K, where f has a bounded continuous inner limit. Thus all poles lie outside K. Finally (5.3) says exactly that $f(\infty) = \infty$. $\square$

Proposition 4 (Full-level identity)**.** *For the reduced rational extremizer $f = U/V$,*

$$\{z \in \mathbb{C} : |f(z)| < 1\} = \Omega, \qquad \{z \in \mathbb{C} : |f(z)| = 1\} = \partial K. \tag{6.3}$$

Moreover f' does not vanish on ∂K, and ∂K is a smooth real-analytic strictly convex Jordan curve.

Proof. The domain Ω is a component of $\mathcal{O} := \{|f| < 1\}$. Indeed, $|f| < 1$ in Ω and $|f| = 1$ on ∂K, so a path in $\mathcal{O}$ cannot leave K. Every component $\mathcal{O}_0$ is bounded because $f(\infty) = \infty$, and it contains a zero of f. Otherwise its compact closure would avoid every zero and pole of f; hence $1/f$ would be holomorphic on a neighborhood of $\overline{\mathcal{O}_0}$, equal in modulus to 1 on the boundary, and strictly larger than 1 inside. Its maximum would occur in the interior, contrary to the maximum-modulus principle. Lemma 10 puts every zero in Ω, so no component other than Ω exists. This proves the first identity.

The boundary of $\mathcal{O}$ is contained in $\{|f| = 1\}$. Conversely, if $|f(z_0)| = 1$, the open mapping theorem says that every neighborhood of z_0 contains points where $|f| < 1$. Thus $z_0 \in \partial\mathcal{O}$, proving the second identity.

If f' vanished to order $k - 1 \ge 1$ at a level point, then a local analytic logarithm of $f/f(z_0)$ would start with $c(z-z_0)^k$. Its zero-real-part set would have $2k \ge 4$ local rays. That is impossible for the boundary of a convex body, which is locally a Jordan arc. Hence $f' \neq 0$ on the level, and the implicit-function theorem makes the boundary real analytic and smooth.

Finally suppose that the boundary contained a nontrivial segment of a real line $z = z_0 + t\xi$, $t \in \mathbb{R}$, $\xi \neq 0$. The expression

$$|U(z_0 + t\xi)|^2 - |V(z_0 + t\xi)|^2$$

is a polynomial in the real variable t. Its vanishing on an interval would make it vanish for every real t. Since coprime one-variable polynomials have no common zero, the full-level identity would then put the entire unbounded line in ∂K, contradicting compactness. Thus the boundary has no nontrivial segment, which for a compact convex body is exactly strict convexity. $\square$

7. The algebraic tangent argument

The Hermitian determinant below is the line-coordinate polynomial introduced by Kippenhahn [6]. We use only projective duality and hyperbolicity.

Proposition 5 (A convex rational lemniscate of numerical-range type is a circle)**.** *Let K be a compact convex body whose boundary is the full level set*

$$\partial K = \{z \in \mathbb{C} : |U(z)/V(z)| = 1\}, \tag{7.1}$$

where U, V are coprime polynomials, $\deg U > \deg V$*, and the level is a smooth strictly convex Jordan curve. Suppose that every tangent line to an open boundary arc belongs to the projective curve*

$$\det(sI + u\mathsf{H} + v\mathsf{J}) = 0 \tag{7.2}$$

for two Hermitian matrices H, J*. Then ∂K is a circle.*

Proof. Write $z = X + iY$, $w = X - iY$, and $S^{\#}(w) = \overline{S(\overline{w})}$ for a polynomial S. For a homogenization $S_h(w, Z)$, the notation $S_h^{\#}$ means coefficientwise conjugation, with w, Z left unchanged. We use primal coordinates $[X : Y : Z]$; a line $sZ + uX + vY = 0$ has dual coordinates $[s : u : v]$. Thus, in vector incidence formulas below, put $\mathbf{z} = (Z, X, Y)^{\mathsf{T}}$ and $\boldsymbol{\ell} = (s, u, v)^{\mathsf{T}}$. Let $d = \deg U$, $e = \deg V < d$, and homogenize the real lemniscate polynomial to degree $2d$:

$$\mathscr{F}(X, Y, Z) = U_h(X + iY, Z)U_h^{\#}(X - iY, Z) - Z^{2(d-e)}V_h(X + iY, Z)V_h^{\#}(X - iY, Z). \tag{7.3}$$

Here U_h, V_h are the usual homogenizations to their own degrees. The boundary lies in the regular locus of $\mathscr{F} = 0$. In fact, at $Z = 1$,

$$\mathscr{F} = |V|^2(|f|^2 - 1), \qquad \partial_z\mathscr{F} = |V|^2 f'\overline{f} \quad \text{on } |f| = 1.$$

Here $V \neq 0$ and $f' \neq 0$ on the level by Proposition 4. The smooth connected curve ∂K therefore lies in one complex irreducible factor $\mathscr{C}$ of $\mathscr{F}$. Locally only one factor can vanish at a regular point, and connectedness keeps the same factor along the boundary. Conjugation preserves this factor because it contains a real arc, so its equation can be chosen real.

Every real affine point of $\mathscr{C}$ is a real zero of (7.3), hence is on the full level (7.1); coprimality prevents U, V from vanishing there simultaneously. Thus

$$\mathscr{C}(\mathbb{R}) = \partial K$$

in the affine plane. There are no additional real points at infinity, because at $Z = 0$

$$\mathscr{F}(X, Y, 0) = |\operatorname{lc}(U)|^2(X^2 + Y^2)^d, \tag{7.4}$$

which has no real projective zero. Therefore the real projective locus of $\mathscr{C}$ is exactly the one oval ∂K.

Let $\mathscr{C}^*$ be the projective dual curve, the Zariski closure of the tangent lines to $\mathscr{C}$, and let $\mathscr{Q}(s, u, v)$ be its irreducible homogeneous equation, chosen with real coefficients. Tangency on an open real arc and (7.2) imply

$$\mathscr{Q}(s, u, v) \text{ divides } \det(sI + u\mathsf{H} + v\mathsf{J}). \tag{7.5}$$

Indeed, that real-analytic arc is Zariski dense in the irreducible curve $\mathscr{C}^*$. The determinant therefore vanishes on all of $\mathscr{C}^*$. Its homogeneous prime ideal is the principal ideal $(\mathscr{Q})$, which proves the divisibility.

Set $\mathbf{e} = [1:0:0]$. The determinant in (7.5) equals 1 at $\mathbf{e}$, so $\mathcal{Q}(\mathbf{e}) \neq 0$. For real u, v, all roots in s of the determinant are real, because they are the negatives of the eigenvalues of the Hermitian matrix $u\mathsf{H} + v\mathsf{J}$. Hence every root of $\mathcal{Q}(s,u,v)$ is real. In other words, $\mathcal{Q}$ is hyperbolic with respect to $\mathbf{e}$.

Let $m = \deg \mathcal{Q}$. Choose a generic real direction $[u:v]$. Since $\mathcal{Q}(\mathbf{e}) \neq 0$, the coefficient of s^m in $\mathcal{Q}(s,u,v)$ is nonzero. Here are the finite exceptional sets being avoided. On the normalization of $\mathscr{C}$, the Gauss map to $\mathscr{C}^*$ is generically one-to-one by projective biduality in characteristic zero. Its ramification locus, the singular locus of $\mathscr{C}^*$, and the tangent images of the finitely many singular points of $\mathscr{C}$ are finite. Avoid their projections to $[u:v]$, as well as the discriminant directions of $\mathcal{Q}(s,u,v)$. The discriminant is not identically zero because the generic projection of an irreducible curve is separable in characteristic zero. The specialization then has m distinct roots, and hyperbolicity makes the corresponding dual points $\ell_j = [s_j : u : v]$ real and smooth.

Biduality gives each ℓ_j a unique contact point on the smooth locus of $\mathscr{C}$. In the standard primal coordinate order this point is

$$[X:Y:Z] = [\mathcal{Q}_u(\ell_j) : \mathcal{Q}_v(\ell_j) : \mathcal{Q}_s(\ell_j)],$$

so it is real. It therefore belongs to $\mathscr{C}(\mathbb{R}) = \partial K$.

A smooth strictly convex oval has exactly two tangent lines with a prescribed unoriented normal $[u:v]$: the support line at the maximum and the one at the minimum of $uX + vY$. The preceding count shows that there are at most two roots; conversely these two support lines belong to $\mathscr{C}^*$, so both are roots. Consequently

$$m = 2. \tag{7.6}$$

Thus $\mathscr{C}^*$ is an irreducible conic and is nonsingular. Writing its equation as $\boldsymbol{\ell}^{\mathsf{T}}\Sigma\boldsymbol{\ell} = 0$, with an invertible symmetric 3×3 matrix Σ, its tangent at $\boldsymbol{\ell}$ corresponds to the primal point $\Sigma\boldsymbol{\ell}$. The envelope of those tangents is $\mathbf{z}^{\mathsf{T}}\Sigma^{-1}\mathbf{z} = 0$, also a conic. Since it contains a Zariski-dense arc of $\mathscr{C}$, it is $\mathscr{C}$.

Finally, because $\mathscr{C}$ divides $\mathscr{F}$, every point of this conic at infinity must be one of

$$[1:i:0], \qquad [1:-i:0].$$

A real irreducible conic has two points at infinity counted with multiplicity, and conjugation exchanges these two circular points. It therefore contains both. Its real homogeneous equation has the form

$$\alpha(X^2+Y^2) + \beta XZ + \gamma YZ + \delta Z^2 = 0, \qquad \alpha \neq 0.$$

Completing squares shows that its nonempty compact real oval is a circle. □

8. Proof of the main theorem

Proof of Theorem 1. We use induction on N. For $N = 1$, and more generally whenever the numerical range has empty interior, Lemma 1 gives $\psi(A) = 1$, so the hypothesis never occurs.

Assume the result below size N, and let $A \in M_N(\mathbb{C})$ satisfy $\psi(A) = 2$. Put $K = W(A)$. It has nonempty interior. By Lemma 6, either K is already a nondegenerate disk or $\operatorname{spec}(A) \subset \Omega = \operatorname{int} K$. Consider the second case.

Lemma 7 supplies a finite-Blaschke extremizer $f \in A(K)$ satisfying (4.6). Proposition 2 supplies x, y and the exact boundary-kernel identity. Since an interior-spectrum numerical range cannot be a polygon, Lemma 8 supplies a curved analytic boundary arc. On that arc, Proposition 3 proves

$$f(z) = \frac{U(z)}{V(z)}, \qquad \deg U > \deg V,$$

after cancellation of the meromorphic transfer identity (5.2).

The variation fe^{-th} makes $x^*h(A)x$ a positive state (Lemma 9). Its Cauchy transform has no zero outside the convex set K, so every zero of the reduced numerator U lies in Ω, while the poles lie outside K. The degree inequality makes f blow up at infinity. Proposition 4 then gives

$$\{|f| < 1\} = \Omega, \qquad \{|f| = 1\} = \partial K,$$

and makes this boundary a smooth strictly convex rational lemniscate.

For $H_A = \operatorname{Re} A$ and $J_A = \operatorname{Im} A$, every supporting tangent line $s + uX + vY = 0$ to $W(A)$ satisfies

$$\det(sI + uH_A + vJ_A) = 0 :$$

the support matrix is semidefinite and has a unit vector attaining the boundary point in its kernel. All hypotheses of Proposition 5 now hold. It follows that ∂K is a circle. Compactness and convexity make K the filled closed disk bounded by that circle.

The disk has positive radius because K has nonempty planar interior. Writing its center and radius as γ and ρ gives

$$W(A) = \{z : |z - \gamma| \leq \rho\}, \qquad \rho > 0,$$

which completes the induction and the proof. $\square$

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